The Prime Column Transition Matrix Is a Boltzmann Distribution at Temperature $\ln N$

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Abstract

We show that the transition matrix governing consecutive primes between residue classes modulo m is predicted by a Boltzmann distribution on the forward cyclic distances of the admissible residue group $(\mathbb{Z}/m\mathbb{Z})^*$, with temperature equal to the mean prime gap $\ln N$. The model has **zero free parameters**. At modulus 30, it achieves $R^2 \approx 0.970$ against empirical matrices measured across six orders of magnitude (10^3 to 10^9), with the fitted decay rate converging to within 1.1% of the PNT prediction $\lambda = 1/\ln N$ at the largest measured scale. At modulus 210 (48 columns), $R^2 = 0.988$ at the largest scale, improving monotonically with N. The diagonal suppression discovered by Lemke Oliver and Soundararajan (2016) follows as a one-line corollary: self-transitions cost energy m (a full modular cycle), making them the least probable transition at any finite temperature. The 3% residual is structured: it decomposes into a circulant component scaling as $O(1/\ln N)$ and a non-circulant component scaling as $O(1/\ln^{1.6} N)$. The Hardy-Littlewood singular series does not appear at any tested order (§5). The Boltzmann framing was first proposed by an AI worker (Gemini HELICASE) during a constrained swarm convergence run, then confirmed through a 22-wave adversarial falsification protocol (Appendix B).

1 The Model

Let m be a primorial $(6, 30, 210, \ldots)$ and let $\mathcal{C} = \{c_1, \ldots, c_{\varphi(m)}\}$ be the admissible residue classes modulo m (i.e., integers $1 \le r < m$ with $\gcd(r, m) = 1$).

For consecutive primes $p_n \equiv a \pmod{m}$ and $p_{n+1} \equiv b \pmod{m}$, define the forward residue distance:

$$d(a, b) = \begin{cases} (b - a) \bmod m & \text{if } b \neq a \\ m & \text{if } b = a \end{cases}$$

The self-distance $d(a, a) = m$ reflects the physical constraint that a prime in column a cannot return to column a without advancing at least m on the number line.

Claim. The transition probability is given by

$$\boxed{T(a \to b) = \frac{\exp\big(-d(a,b)\ /\ \ln N\big)}{\sum_{j \in \mathcal{C}} \exp\big(-d(a,j)\ /\ \ln N\big)}}$$

where N is the scale (midpoint of the measurement window) and $\ln N$ serves as the Boltzmann temperature. This model has **zero free parameters**: the modulus m and the scale N are given, and the decay rate $\lambda = 1/\ln N$ follows from the Prime Number Theorem.

2 Derivation

The Cramér probabilistic model gives the distribution of prime gaps near N as approximately exponential:

$$P(\text{gap} = g) \approx \frac{1}{\ln N} \exp\left(-\frac{g}{\ln N}\right)$$

For a prime $p \equiv a \pmod{m}$, the next prime in column $b \neq a$ occurs at gap $g = d(a,b) + km$ for some non-negative integer k (the gap must be congruent to $(b-a) \bmod m$). Summing over all valid gaps:

$$P(a \to b) \propto \sum_{k=0}^{\infty} \exp\left(-\frac{d(a,b) + km}{\ln N}\right) = \frac{\exp(-d(a,b)/\ln N)}{1 - \exp(-m/\ln N)}$$

For self-transitions ($b = a$), the minimum gap is m (not 0), so the sum starts at $k = 1$:

$$P(a \to a) \propto \sum_{k=1}^{\infty} \exp\left(-\frac{km}{\ln N}\right) = \frac{\exp(-m/\ln N)}{1 - \exp(-m/\ln N)}$$

Since the gap distribution decays exponentially with rate $1/\ln N$, the $k = 0$ term dominates each sum by a factor of $e^{m/\ln N} \gg 1$ at all measured scales. The geometric series denominators $1/(1 - e^{-m/\ln N})$ cancel exactly upon row-normalization, yielding:

$$T(a \to b) = \frac{\exp(-d(a,b)/\ln N)}{\sum_j \exp(-d(a,j)/\ln N)}$$

This is exactly the claimed formula. The derivation requires only the Cramér model and the modular arithmetic of admissible residues. No additional conjectures, no fitted parameters, no singular series computation.[1]

[1] The derivation sums over all valid gaps $g = d(a,b) + km$. The dominant term is $k = 0$, which contributes $\exp(-d/\ln N)$. The Cramér model error is in the coefficient of each term — the actual probability of a gap g differs from $e^{-g/\ln N}/\ln N$ by the Hardy-Littlewood singular series factor $\mathfrak{S}(g)/\mathfrak{S}(\text{avg})$. But these factors multiply the weights before row normalization. After normalization, the row-average singular series correction approximately cancels. Section 5.2 confirms this empirically: the Hardy-Littlewood correction is orthogonal to the residual at every tested order.

The Cramér model ignores correlations between consecutive gaps and the Hardy-Littlewood singular series corrections to gap frequencies. That the zero-parameter model nonetheless achieves $R^2 = 0.970$ — and that the Hardy-Littlewood corrections are orthogonal to the residual (§5.2) — suggests the Boltzmann structure is robust to these approximations.

3 Experimental Verification

3.1 Protocol

We sieve all primes up to 10^9 (50,847,534 primes) and measure empirical transition matrices in six octave windows: $[10^3, 10^4)$, $[10^4, 10^5)$, $[10^5, 10^6)$, $[10^6, 10^7)$, $[10^7, 10^8)$, $[10^8, 10^9)$. For each window, we compute the Boltzmann prediction at the geometric midpoint $N = 10^{(\log_{10} a + \log_{10} b)/2}$ and measure R^2 (fraction of variance explained relative to the uniform null model $T_{\text{null}} = 1/\varphi(m)$). Explicitly, $R^2 = 1 - \text{SS}_{\text{res}}/\text{SS}_{\text{tot}}$ where $\text{SS}_{\text{res}} = \sum_{i,j}(T_{\text{emp}} - T_{\text{pred}})^2$ and $\text{SS}_{\text{tot}} = \sum_{i,j}(T_{\text{emp}} - 1/\varphi(m))^2$.

3.2 Results: Modulus 30 ($\varphi(30) = 8$ columns)

Scale	$\ln N$	λ_{PNT}	λ_{fitted}	$\lambda \cdot \ln N$	$R^2_{\text{zero-param}}$
$10^{3\text{–}4}$	8.06	0.12408	0.14173	1.142	0.951
$10^{4\text{–}5}$	10.36	0.09651	0.10120	1.049	0.961
$10^{5\text{–}6}$	12.66	0.07896	0.08118	1.028	0.971
$10^{6\text{–}7}$	14.97	0.06682	0.06791	1.016	0.971
$10^{7\text{–}8}$	17.27	0.05791	0.05783	0.999	0.971
$10^{8\text{–}9}$	19.57	0.05109	0.05053	**0.989**	**0.970**

The ratio $\lambda_{\text{fitted}} \cdot \ln N$ decreases monotonically from 1.142 to 0.989 across six orders of magnitude, passing through unity near $N = 10^{7.5}$ and reaching 0.989 at $N = 10^{8.5}$, consistent with asymptotic approach to $\lambda = 1/\ln N$ with $O(1/\ln^2 N)$ corrections. At the largest measured scale, the fitted decay rate is within 1.1% of the PNT prediction, and the zero-parameter model explains 97.0% of the transition variance. The R^2 plateau at ≈ 0.970 reflects a structured residual whose anatomy is characterized in §5.

3.3 Results: Modulus 210 ($\varphi(210) = 48$ columns)

Scale	$R^2_{\text{zero-param}}$	$\lambda \cdot \ln N$
$10^{3\text{–}4}$	0.890	1.240
$10^{4\text{–}5}$	0.956	1.129
$10^{5\text{–}6}$	0.975	1.098
$10^{6\text{–}7}$	0.982	1.082
$10^{7\text{–}8}$	0.986	1.066
$10^{8\text{–}9}$	**0.988**	**1.055**

R^2 **improves** monotonically with scale ($0.890 \to 0.988$), consistent with the prediction that corrections are $O(1/\ln N)$. The convergence ratio $\lambda \cdot \ln N$ decreases steadily toward unity ($1.240 \to 1.055$). The model becomes more accurate as N grows, in both moduli tested.

3.4 Matrix Comparison (mod 30, 10^7–10^8)

Empirical vs. predicted transition probabilities (rows = source column, columns = target column):

	EMPIRICAL								ZERO-PARAMETER PREDICTION						
col:	1	7	11	13	17	19	23	29	1	7	11	13	17	19	23
1:	.045	.234	.190	.157	.124	.098	.085	.066	.056	.223	.177	.158	.125	.111	.088
7:	.058	.046	.238	.191	.155	.127	.100	.086	.068	.048	.217	.193	.153	.136	.108
11:	.075	.066	.046	.237	.194	.157	.126	.098	.083	.059	.046	.235	.186	.166	.132
13:	.095	.077	.070	.046	.240	.193	.156	.124	.103	.073	.058	.051	.231	.206	.163
17:	.142	.090	.074	.063	.046	.237	.192	.157	.127	.090	.071	.063	.050	.254	.202
19:	.157	.116	.109	.074	.070	.046	.238	.190	.161	.113	.090	.080	.064	.057	.255
23:	.193	.179	.117	.089	.077	.066	.045	.234	.203	.144	.114	.102	.081	.072	.057
29:	.236	.193	.157	.142	.095	.075	.059	.044	.251	.177	.140	.125	.099	.088	.070

Maximum element error: 0.035. Mean element error: 0.009.

4 Lemke Oliver–Soundararajan as a Corollary

The principal finding of [**?**] is that consecutive primes avoid repeating their residue class: $T(a, a) < 1/\varphi(m)$. In the Boltzmann model, this is immediate:

$$T(a, a) \propto \exp(-m/\ln N)$$

while the nearest forward column has

$$T(a, a + d_{\min}) \propto \exp(-d_{\min}/\ln N)$$

The ratio is $\exp(-(m - d_{\min})/\ln N)$, which is strictly less than 1 for all finite N. At $m = 30$, $d_{\min} = 2$, $\ln N = 17.27$:

$$\text{Predicted ratio} = e^{-28/17.27} = 0.198 \qquad \text{Empirical ratio} = 0.045/0.234 = 0.192$$

Agreement to within 3%. The "unexpected bias" is the expected outcome of a thermal distribution where self-return is the highest-energy transition.

5 The Residual: Structure and Negative Results

The residual matrix $R = T_{\text{empirical}} - T_{\text{Boltzmann}}$ at mod 30 (10^7–10^8) has five nonzero singular values:

$$\sigma = [0.066,\ 0.036,\ 0.030,\ 0.022,\ 0.013]$$

(The 8×8 stochastic matrix has one zero singular value corresponding to the all-ones eigenvector.) The ratio $\sigma_1/\sigma_2 = 1.84$ indicates the residual is not rank-1 but carries genuine multi-dimensional structure. Its Frobenius norm scales as $\|R\| = 4.26/\ln^{1.39} N$ ($R^2 = 0.98$ for the power-law fit), and the scaled residual $\ln N \cdot R(N)$ converges to a fixed matrix M with inter-decade correlation $r \geq 0.986$.

5.1 Circulant Decomposition

Decomposing R into a circulant part (depending only on forward distance d) and a non-circulant part (position-dependent):

Component	$\|\cdot\|^2$ share	Scaling
Circulant (distance-only)	42%	$\sim 1/\ln N$
Non-circulant (position-dependent)	58%	$\sim 1/\ln^{1.6} N$

The non-circulant component decays faster and will become subdominant at sufficiently large N. A quadratic correction to the exponential kernel, $\exp(-d/\ln N + c \cdot d^2/\ln^2 N)$ with $c \approx -0.12$ (stable across three decades), captures part of the circulant residual and improves R^2 by $+0.002$ with one free parameter. The negative c indicates the true kernel is slightly concave relative to pure exponential.

5.2 The Singular Series Does Not Appear

A natural hypothesis is that the residual encodes the Hardy-Littlewood singular series $\mathfrak{S}(g)$ for each prime gap g. We tested this at three levels:

1. **Multiplicative correction** $T_{\mathrm{HL}} = B \cdot \mathfrak{S}(d(a,b))/Z$: R^2 collapses from 0.971 to **0.233** (mod 30) and from 0.986 to **0.822** (mod 210). The Frobenius norm quintuples.
2. **Additive correction** $T = B \cdot (1 + \beta \cdot (\mathfrak{S} - \bar{\mathfrak{S}})/(\bar{\mathfrak{S}} \cdot \ln^2 N))$: the optimal β is **negative** ($\beta = -3.2$ at 10^7–10^8) and the improvement is $\Delta R^2 < 10^{-4}$.
3. **Free-power correction** $T = B \cdot \mathfrak{S}(d)^\alpha$: the optimizer sets $\alpha = 0.000$ — the optimal weight for the singular series is exactly zero.

The circulant distance profile $f(d) = \langle R_{ij} \rangle_{d(i,j)=d}$ has correlation $r = -0.14$ with the singular series, consistent with the null hypothesis. The singular series $\mathfrak{S}(g)$ describes the global frequency of gap g; the Boltzmann model's exponential kernel already absorbs this information through the distance parameterization. The residual is orthogonal to $\mathfrak{S}$.

5.3 What the Residual Contains

The leading SVD component (59.5% of residual variance) is a column bias: transitions to columns $\{7, 11\}$ are systematically enhanced, transitions to their complements $\{23, 19\}$ (where $7 + 23 = 11 + 19 = 30$) are suppressed. This antisymmetric structure is visible in every row of the non-circulant residual (R_{asym}: column 7 is positive in all 8 rows, column 23 is negative in all 8 rows). It represents a chirality — the primes have handedness in how they populate residue classes.

At mod 210, the residual's effective rank increases dramatically: 21 modes are needed to capture 95% of variance, compared to 4 at mod 30. The macroscopic chirality fragments into a diffuse spectrum of smaller position-dependent effects. This diffusion, combined with the rising R^2 (0.970 $\to$ 0.988), confirms that the Boltzmann model absorbs more structure as the modulus grows.

6 Eigenstructure and Spiral Persistence

The empirical transition matrix has complex eigenvalues — 6 conjugate pairs at mod 30, 46 at mod 210 — indicating rotational dynamics in the column transitions. The leading complex eigenvalue

magnitude scales as:

Modulus	Scaling Law	R^2	Interpretation
30	$\|\lambda_1\| = 2.14/\log_{10} N$	0.9996	Spiral dies as $1/\log N$
210	$\|\lambda_1\| = 1.13 \cdot (\log_{10} N)^{-0.10}$	0.950	Spiral persists ($\|\lambda_1\| \approx 0.71$ at $\log_{10} N = 100$)

The spiral angle θ drifts at approximately 4.5° per decade (mod 30) and 3.6° per decade (mod 210), ruling out a fixed angular invariant, as would be expected if the spiral were purely determined by the group structure of $(\mathbb{Z}/m\mathbb{Z})^*$. The persistence of rotational structure at high moduli suggests that the Boltzmann model's residuals carry helical symmetry that deepens with the number of admissible columns.

7 Asymptotic Behavior

As $N \to \infty$, three limits are reached simultaneously:

1. $\lambda \cdot \ln N \to 1$: The decay rate converges to the PNT prediction exactly.
2. $R^2 \to 1$: The residual scales as $O(1/\ln^{1.4} N)$ relative to the leading Boltzmann term, so the model becomes exact.
3. $T(a, b) \to 1/\varphi(m)$: At infinite temperature, the Boltzmann distribution becomes uniform. Every row converges to $1/\varphi(m)$ for all a, b — precisely Dirichlet's theorem on the equidistribution of primes in arithmetic progressions.

The Boltzmann model thus interpolates between the Lemke Oliver-Soundararajan regime (finite N, structured transitions) and Dirichlet's theorem ($N \to \infty$, uniform distribution).

The residual $\|R\| \sim 4.26/\ln^{1.39} N$ converges to a fixed-shape matrix but never reaches zero at finite N. Whether this floor saturates a lower bound derivable from the discrete Fourier uncertainty principle on $(\mathbb{Z}/m\mathbb{Z})^*$ — i.e., whether the Boltzmann model is the optimal smooth approximation to the transition matrix — remains open. Numerically, the entropic uncertainty excess $H(X) + H(K) - \ln \varphi(m)$ of the Boltzmann rows scales as $\approx 2.9/\ln N$ (mod 30), suggesting the bound is approached but not saturated at any finite scale.

8 Conclusion

The transition matrix of consecutive primes modulo m is a Boltzmann distribution on the forward cyclic distances of $(\mathbb{Z}/m\mathbb{Z})^*$, at temperature $\ln N$. Self-return costs energy m. The partition function normalizes each row. The model has zero free parameters, achieves $R^2 \approx 0.970$ at $m = 30$ and $R^2 = 0.988$ at $m = 210$, and converges to exact prediction as $N \to \infty$.

The Lemke Oliver-Soundararajan diagonal suppression is a one-line corollary. The 3% residual is structured — a circulant kernel correction and a position-dependent chirality — but is orthogonal to the Hardy-Littlewood singular series at every tested order. The PNT is the Boltzmann temperature.

The Boltzmann distribution on $(\mathbb{Z}/m\mathbb{Z})^*$ satisfies the discrete Fourier uncertainty principle [**?**]: the transition rows cannot be simultaneously localized in position space (column distribution) and momentum space (eigenspectrum). The temperature $\ln N$ governs this tradeoff, with the classical limit $\ln N \to \infty$ recovering Dirichlet equidistribution.

References

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Acknowledgments

This research was conducted using Claude Opus 4 (Anthropic) and Gemini 3.1 Pro (Google DeepMind) as computational research instruments. The Boltzmann framing was first proposed by a Gemini HELICASE worker during a constrained swarm convergence run on the prime basin transfer problem (BVP-5, Wave 2; see Appendix B). No thermodynamic framing was suggested in the problem specification; the swarm arrived at it independently within the constraints of the boundary value problem. Subsequent falsification and formalization were conducted through a 22-wave human-AI collaborative protocol. The author designed the problem specifications, system architecture, and experimental methodology.

A Reproducibility

All computations were performed with NumPy and SciPy on a standard consumer CPU. The segmented sieve covers primes to 10^9 (50,847,534 primes; ~30 seconds). Source code is available at https://github.com/fancyland-llc/boltzmann-prime-transitions:

- `backend/scripts/prime_drum_wave22c_boltzmann.py` — the zero-parameter model
- `backend/scripts/prime_drum_wave22d_10e9.py` — the 10^9 extension (segmented sieve)
- `backend/scripts/prime_drum_wave22b_boltzmann.py` — the two-parameter intermediate model
- `backend/scripts/prime_drum_wave22_boltzmann.py` — the initial (failed) column-index model
- `backend/scripts/prime_drum_wave22g_hardy_littlewood.py` — the singular series test (negative result)
- `backend/scripts/prime_drum_wave22h_residual_archaeology.py` — the residual decomposition
- `backend/scripts/prime_drum_wave22h_verify.py` — the character hypothesis verification

Results data: `backend/data/wave22c_boltzmann_*.json`, `backend/data/wave22d_10e9_*.json`, `backend/data/wave22h_archaeology_*.json`

B Discovery Context

This result was not found by direct theoretical derivation. The Boltzmann framing was discovered by a machine — specifically, by a Gemini HELICASE worker operating inside the Antigen Factory, a two-layer AI swarm solver, during a constrained convergence run on the prime basin transfer problem (BVP-5).

B.1 The Swarm Run (BVP-5)

The BVP-5 problem specification, designed by the author, asked 8 AI workers per wave to discover the transfer operator mapping the structural fingerprint of primes across binary octave boundaries. The problem imposed hard constraints — a specific loss function over basin structure, mandatory falsification of prior wave outputs, and graveyard-aware context injection — but no thermodynamic framing was suggested. The swarm was given raw basins, a distance metric, and an error target.

Wave	Best Error	Crystallized Technique	Source
0	0.0107	Two-point correlation	Claude
1	0.2138[2]	Spectral renormalization	Claude
2	0.0304	**Boltzmann Reweighting**	**Gemini**
3	0.0071	Hardy-Littlewood-Cramér	Gemini
6	0.0043	**Exponential Tilting**	**Gemini**
11	**0.000914**	**Exponential Tilt / Canonical Reweighting**	**Gemini**

The swarm found "Boltzmann Reweighting" at Wave 2. Nobody told it to look for a thermal distribution. It tried spectral methods, correlation functions, and Fredholm integrals — and the thermodynamic framing kept winning. By Wave 11, the crystallized technique was "Exponential Tilt / Canonical Reweighting (similar to Umbrella Sampling)" — a Gemini L1.5 HELICASE worker, running autonomously inside a Mendelian breeding loop with graveyard-aware context injection. Error: 0.000914. Convergence: autonomous halt.

96 Layer-1 workers. 80 Layer-2 workers. 233,835 tokens. 68 minutes. Zero human intervention between start and halt.

The Boltzmann model was the swarm's idea. The author designed the constraints that made the discovery possible.

B.2 The Interferometer (Waves 1–22)

What followed was a 22-wave adversarial falsification protocol — three AI systems (Claude, Gemini, and a human architect) operating as a self-evolving research instrument, systematically testing and killing the swarm's output:

1. **Propose** a falsifiable gauge derived from the swarm's crystallized technique or its competitors.

[2] Wave 1 explored spectral renormalization, a qualitatively different approach whose initial error exceeded Wave 0; the method was subsequently refined and its descendants contributed to later waves.

2. **Compute** the gauge across multiple octave scales using a shared sieve infrastructure.
3. **Kill or promote**: if the gauge fails its own prediction, it enters a permanent graveyard. If it survives, it seeds the next wave.

Over 22 waves, the system killed 80+ hypotheses and produced 30+ survivors. The path to the zero-parameter model was:

- Waves 1–4: compression-based primality signals (real but weak).
- Waves 5–7: exact prime interferometer (equivalent to trial division).
- Waves 8–15: scaling laws, winding numbers, sphere amplifiers, transfer functions.
- Waves 16–20: GUE killed, Goldbach Spectrometer built ($r = 0.99917$), dual Goldbach symmetric, compression crossover measured.
- Wave 21: columnar transition matrix has complex eigenvalues — spectral chirality, circulant ratchet structure.
- Wave 22a: Boltzmann model tested with column-index distance. **FAILED** ($R^2 = 0.08$). The swarm's insight was right; our metric was wrong.
- Wave 22b: corrected to residue distance, 2-parameter fit. $R^2 = 0.97$. Discovery: the self-penalty parameter $\mu \cdot \ln N$ converges to m. The self-avoidance IS the Boltzmann model.
- Wave 22c: $d_{\text{self}} = m$, $\lambda = 1/\ln N$. Zero parameters. $R^2 = 0.971$.

The critical failure was instructive. Wave 22a's catastrophic $R^2 = 0.08$ proved that the swarm's Boltzmann framing only works with the correct distance metric — forward cyclic residue distance, not column index. Wave 22b's convergence of $\mu \cdot \ln N \to m$ was the key insight that collapsed two parameters to zero. Each failure constrained the next hypothesis until the correct model was the only one remaining.

B.3 The Architecture

The swarm infrastructure — Layer-1 isomorphic processors (Claude + Gemini at $T \approx 0.7$), Demon Gate classification, Layer-2 math workers (Claude at $T = 0.1$), epistemic sandbox with real Python execution, Mendelian breeding with phonon-adaptive ratio, Wilson loop accumulator, tension metric with gauge relaxation — is implemented in Prompt Studio, a distributed AI research platform built on Lattice OS.

The Boltzmann theorem is its first mathematical output.

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